BIF604 Comprehensive MCQs (Lessons 56 - 137)

Section 1: Bioinformatics Tools (DAVID, STRING, Cytoscape)

Lesson 56 - 63: DAVID & Functional Annotation

1. **What does DAVID stand for?**
   - A) Database for Annotation, Visualization and Integrated Discovery
   - B) Digital Annotation and Visualization for Integrated Data
   - C) Database for Analysis, Visualization and Integrated Development
   - D) Digital Analysis for Visualization and Integrated Discovery
   - **Answer: A**

2. **DAVID is primarily used for the functional interpretation of large datasets derived from:**
   - A) Protein sequencing
   - B) Microarray and other genomic studies
   - C) X-ray crystallography
   - D) NMR spectroscopy
   - **Answer: B**

3. **Which of the following is NOT a tool suite embedded in DAVID?**
   - A) Gene functional classification Tool
   - B) Gene ID conversion Tool
   - C) NIAID Pathogen Genome Browser
   - D) Protein Structure Prediction Tool
   - **Answer: D**

4. **The Gene Ontology (GO) terms are grouped into which three categories?**
   - A) Molecular function, Biological process, Cellular component
   - B) DNA, RNA, Protein
   - C) Genotype, Phenotype, Haplotype

- D) Transcription, Translation, Replication
- **Answer: A**

5. **Which category of GO describes the location in the cell where a gene product executes its function?**
   - A) Molecular function
   - B) Biological process
   - C) Cellular component
   - D) Genomic context
   - **Answer: C**

6. **In DAVID, the "Functional Annotation Tool" highlights relevant GO terms associated with a gene list using a:**
   - A) Clustering algorithm
   - B) Novel algorithm to measure co-association relationships
   - C) BLAST search
   - D) Sequence alignment
   - **Answer: B**

7. **KEGG stands for:**
   - A) Kyoto Encyclopedia of Genes and Genomes
   - B) Knowledge Entry for Genes and Genomes
   - C) Key Encyclopedia of Genomic Groups
   - D) Kyoto Entry for Gene Groups
   - **Answer: A**

Lesson 64 - 68: Gene Networks & STRING

1. **In a gene interaction network, nodes represent _____ and edges represent _____.**
   - A) Interactions, Genes
   - B) Genes, Interactions
   - C) Proteins, DNA
   - D) mRNA, Proteins
   - **Answer: B**

2. **What is the “Degree” of a network node?**
    - A) The size of the node
    - B) The number of edges connected to a node
    - C) The color of the node
    - D) The intensity of gene expression
    - **Answer: B**

3. **STRING is a database of known and predicted:**
    - A) DNA sequences
    - B) Protein-protein interactions
    - C) RNA structures
    - D) Metabolic pathways
    - **Answer: B**

4. **Interaction scores in STRING indicate:**
    - A) The strength of the interaction
    - B) The confidence that an interaction is true
    - C) The speed of the interaction
    - D) The physical distance between proteins
    - **Answer: B**

5. **In STRING’s “Evidence Mode,” an edge can be drawn with up to how many differently colored lines?**
    - A) 3
    - B) 5
    - C) 7
    - D) 10
    - **Answer: C**

6. **Which of the following is NOT a source of data for STRING?**
    - A) Genomic context predictions
    - B) High-throughput lab experiments
    - C) Automated text-mining
    - D) Real-time weather data

- **Answer: D**

Lesson 69 - 76: Cytoscape & Network Analysis

1. **Cytoscape is an open-source software platform for:**
   - A) Writing Python code
   - B) Visualizing and integrating molecular interaction networks
   - C) Sequencing DNA
   - D) Predicting protein folding
   - **Answer: B**

2. **Which menu in Cytoscape contains options for creating new networks and importing data?**
   - A) Edit
   - B) View
   - C) File
   - D) Apps
   - **Answer: C**

3. **How many ways are there to create networks in Cytoscape?**
   - A) 2
   - B) 3
   - C) 4
   - D) 5
   - **Answer: C**

4. **Which Cytoscape layout algorithm places nodes in a circular arrangement based on attributes?**
   - A) Grid Layout
   - B) Attribute Circle Layout
   - C) Hierarchical Layout
   - D) Spring-Embedded Layout
   - **Answer: B**

5. **The “BiNGO” plugin in Cytoscape is used to:**

- A) Predict protein structures
- B) Determine statistically overrepresented GO categories
- C) Perform sequence alignment
- D) Edit DNA sequences
- **Answer: B**

6. **What is a “Style” in Cytoscape?**
   - A) A way to write code
   - B) A set of visual mappings (color, size, shape) for network data
   - C) A type of network file format
   - D) A statistical test
   - **Answer: B**

7. **NetworkAnalyzer in Cytoscape can calculate:**
   - A) Node Degree Distribution
   - B) Betweenness Centrality
   - C) Clustering Coefficient
   - D) All of the above
   - **Answer: D**

Section 2: Python Programming (Lessons 77 - 98)

Lesson 77 - 82: Python Basics & Strings

1. **Which of the following is a high-level programming language?**
   - A) Assembly
   - B) Machine Code
   - C) Python
   - D) Binary
   - **Answer: C**

2. **In Python, comments start with which character?**
   - A) //

- B) /*
- C) #
- D) –
- **Answer: C**

3. **What is the output of `print(9/10)` in Python 3?**
   - A) 0
   - B) 0.9
   - C) 1
   - D) Error
   - **Answer: B**

4. **Which operator is used for exponentiation in Python?**
   - A) ^
   - B) *
   - C) **
   - D) //
   - **Answer: C**

5. **What does the `len()` function return for a string?**
   - A) The number of words
   - B) The number of characters
   - C) The memory size
   - D) The first character
   - **Answer: B**

6. **Which method removes whitespace from both ends of a string?**
   - A) `strip()`
   - B) `rstrip()`
   - C) `lstrip()`
   - D) `remove()`
   - **Answer: A**

7. **What is the result of `'ATGC'.find('G')`?**

- A) 1
- B) 2
- C) 3
- D) 0
- **Answer: B**

Lesson 83 - 86: Loops & Lists

1. **Which loop is used when the number of iterations is known?**
   - A) while
   - B) for
   - C) do-while
   - D) if
   - **Answer: B**
2. **The `range(5)` function generates numbers from:**
   - A) 1 to 5
   - B) 0 to 4
   - C) 0 to 5
   - D) 1 to 4
   - **Answer: B**
3. **Which list method adds an element to the end of the list?**
   - A) `add()`
   - B) `insert()`
   - C) `append()`
   - D) `extend()`
   - **Answer: C**
4. **Lists in Python are:**
   - A) Immutable
   - B) Mutable
   - C) Fixed size
   - D) Unordered

- **Answer: B**

5. **What is the result of `[1, 2] + [3, 4]`?**
   - A) `[4, 6]`
   - B) `[1, 2, 3, 4]`
   - C) `[[1, 2], [3, 4]]`
   - D) Error
   - **Answer: B**

Lesson 87 - 93: Files, Tuples, Dictionaries, Sets

1. **Which mode is used to open a file for writing, deleting previous content?**
   - A) ‘r’
   - B) ‘a’
   - C) ‘w’
   - D) ‘x’
   - **Answer: C**

2. **Which function is used to open a file in Python?**
   - A) `file()`
   - B) `open()`
   - C) `read()`
   - D) `load()`
   - **Answer: B**

3. **Tuples are defined using which brackets?**
   - A) `[]`
   - B) `{}`
   - C) `()`
   - D) `<>`
   - **Answer: C**

4. **Which of the following is TRUE about Tuples?**
   - A) They are mutable

- B) They are immutable
- C) They use curly braces
- D) They cannot be nested
- **Answer: B**

5. **Dictionaries are collections of:**
   - A) Ordered items
   - B) Key-value pairs
   - C) Unique integers
   - D) Only strings
   - **Answer: B**

6. **In a dictionary, keys must be:**
   - A) Mutable
   - B) Immutable
   - C) Lists
   - D) Any type
   - **Answer: B**

7. **Which set operation returns elements present in both sets?**
   - A) Union (|)
   - B) Intersection (&)
   - C) Difference (-)
   - D) Symmetric Difference (^)
   - **Answer: B**

Lesson 94 - 98: Modules, Classes, Exceptions

1. **A Python file with a `.py` extension is called a:**
   - A) Package
   - B) Module
   - C) Class
   - D) Function
   - **Answer: B**

2. **To import a specific function `sqrt` from the `math` module, use:**
   - A) `import math.sqrt`
   - B) `from math import sqrt`
   - C) `import sqrt from math`
   - D) `load math(sqrt)`
   - **Answer: B**

3. **A directory must contain which file to be recognized as a Python package?**
   - A) `main.py`
   - B) `__init__.py`
   - C) `setup.py`
   - D) `package.py`
   - **Answer: B**

4. **In Python OOP, `self` represents:**
   - A) The class itself
   - B) The instance of the class
   - C) A global variable
   - D) A private method
   - **Answer: B**

5. **The special method used as a constructor in Python is:**
   - A) `__construct__`
   - B) `__init__`
   - C) `__new__`
   - D) `__start__`
   - **Answer: B**

6. **Which block is used to catch and handle exceptions?**
   - A) `try...catch`
   - B) `try...except`
   - C) `check...handle`
   - D) `error...fix`

- **Answer: B**

Section 3: BioPython (Lessons 99 - 103)

1. **What is the primary purpose of the Biopython project?**
   - A) To provide web development tools
   - B) To provide tools for computational molecular biology
   - C) To create games
   - D) To manage databases
   - **Answer: B**

2. **Which module in Biopython is used to represent sequences?**
   - A) `Bio.Seq`
   - B) `Bio.DNA`
   - C) `Bio.Protein`
   - D) `Bio.Align`
   - **Answer: A**

3. **Which method of a `Seq` object returns the reverse complement?**
   - A) `reverse()`
   - B) `complement()`
   - C) `reverse_complement()`
   - D) `invert()`
   - **Answer: C**

4. **Biopython can parse which of the following file formats?**
   - A) FASTA
   - B) GenBank
   - C) BLAST output
   - D) All of the above
   - **Answer: D**

5. **The `alphabet` object in a Biopython `Seq` object specifies:**
   - A) The font used

- B) The type of sequence (DNA, RNA, Protein)
- C) The language of the code
- D) The file format
- **Answer: B**

Section 4: Phylogenetics (Lessons 111 - 121)

1. **Phylogenetics is the study of:**
   - A) Cell structures
   - B) Evolutionary relationships
   - C) Chemical reactions
   - D) Weather patterns
   - **Answer: B**

2. **In a phylogenetic tree, “Internal Nodes” represent:**
   - A) Extinct species
   - B) Ancestral species
   - C) Existing species
   - D) Individual genes
   - **Answer: B**

3. **What is a “Leaf” in a phylogenetic tree?**
   - A) An ancestral node
   - B) A terminal unit (species, gene, or population)
   - C) A branch length
   - D) The root
   - **Answer: B**

4. **A “Rooted Tree” defines:**
   - A) Only relationships
   - B) A unique evolutionary path from a common ancestor
   - C) No time direction
   - D) Multiple ancestors

- **Answer: B**

5. **Which type of tree depicts only the branching order?**
   - A) Phylogram
   - B) Cladogram
   - C) Ultrametric tree
   - D) Dendrogram
   - **Answer: B**

6. **In a “Phylogram,” branch lengths are proportional to:**
   - A) Time only
   - B) Some measure of divergence
   - C) Number of species
   - D) Random values
   - **Answer: B**

7. **Which method of tree reconstruction is also known as the “minimum evolution method”?**
   - A) Maximum Likelihood
   - B) Maximum Parsimony
   - C) UPGMA
   - D) Neighbor-Joining
   - **Answer: B**

8. **“Informative sites” in Maximum Parsimony must have at least:**
   - A) 1 different nucleotide
   - B) 2 different nucleotides, each present at least twice
   - C) 4 different nucleotides
   - D) 2 mutations
   - **Answer: B**

9. **The Maximum Likelihood (ML) method finds the tree that:**
   - A) Is the shortest
   - B) Maximizes the probability of the data given the tree

- C) Has the fewest branches
- D) Uses the most memory
- **Answer: B**

10. **UPGMA assumes a constant mutation rate, also known as the:**
    - A) Genetic drift
    - B) Molecular clock hypothesis
    - C) Natural selection
    - D) Central dogma
    - **Answer: B**

11. **Which distance-based method does NOT assume a constant mutation rate?**
    - A) UPGMA
    - B) Fitch-Margoliash
    - C) Maximum Parsimony
    - D) ClustalW
    - **Answer: B**

Section 5: Microarray Technology & Analysis (Lessons 122 - 131)

1. **The underlying principle of microarray technology is:**
    - A) Protein folding
    - B) DNA binding to itself or RNA (Hybridization)
    - C) Enzyme catalysis
    - D) Cell division
    - **Answer: B**

2. **What does a “snap-shot” of mRNA isolated from a cell represent?**
    - A) The genome
    - B) The transcriptome
    - C) The proteome
    - D) The metabolome
    - **Answer: B**

3. **Which type of microarray uses 25-nucleotide long probes and photolithography?**
    - A) cDNA array
    - B) Affymetrix
    - C) Nylon membrane
    - D) Macroarray
    - **Answer: B**

4. **In a Two-colour microarray, Cy3 and Cy5 are:**
    - A) Enzymes
    - B) Fluorescent dyes
    - C) Genes
    - D) Probes
    - **Answer: B**

5. **Image processing for microarrays involves identifying which pixels?**
    - A) Foreground (feature) and Background
    - B) Only foreground
    - C) Only background
    - D) Random pixels
    - **Answer: A**

6. **Which file format is typical for scanned microarray data?**
    - A) .exe
    - B) .gpr
    - C) .pdf
    - D) .mp3
    - **Answer: B**

7. **Normalization in microarrays is used to remove:**
    - A) Genes
    - B) Systematic experimental errors and bias
    - C) All expression
    - D) DNA

- **Answer: B**

8. **“Log2 Transformation” of Cy5/Cy3 ratio is used because:**
   - A) It’s easier to multiply
   - B) It treats up-regulation and down-regulation symmetrically
   - C) It removes all noise
   - D) It’s a requirement for Python
   - **Answer: B**

9. **In a HeatMap, red usually represents _____ and green represents _____.**
   - A) Down-regulated, Up-regulated
   - B) Up-regulated, Down-regulated
   - C) No change, Up-regulated
   - D) Mutation, Normal
   - **Answer: B**

10. **Which normalization method makes the distributions of data on all arrays identical?**
    - A) Centering
    - B) Scaling
    - C) Quantile normalization
    - D) Log transformation
    - **Answer: C**

Section 6: Statistics & Cancer Genomics (Lessons 132 - 137)

1. **The Student’s t-distribution is centered at:**
   - A) 1
   - B) 0
   - C) 100
   - D) Alpha
   - **Answer: B**

2. **A p-value < 0.05 generally indicates:**
   - A) The result is not significant

- B) The result is statistically significant
- C) The experiment failed
- D) Too many samples
- **Answer: B**

3. **Which test is used to compare gene expression across two independent groups?**
   - A) One-sample t-test
   - B) Unpaired (Two-sample) t-test
   - C) Paired t-test
   - D) Binomial test
   - **Answer: B**

4. **When should a “Paired t-test” be used?**
   - A) For independent samples
   - B) When observations are correlated (e.g., before and after treatment)
   - C) For very large samples
   - D) For categorical data
   - **Answer: B**

5. **The “Multiplicity Issue” refers to:**
   - A) Having too many genes
   - B) The increased chance of false positives when performing many tests
   - C) Multiple mutations in one gene
   - D) Using multiple computers
   - **Answer: B**

6. **Bonferroni correction controls the multiplicity issue by:**
   - A) Increasing alpha
   - B) Setting alpha to desired alpha divided by number of tests
   - C) Removing genes
   - D) Using a different p-value
   - **Answer: B**

7. **FDR stands for:**

- A) Fast Data Retrieval
- B) False Discovery Rate
- C) Functional Data Reconstruction
- D) Frequency Distribution Rate
- **Answer: B**

8. **cBioPortal is an open-access resource for:**
   - A) Weather data
   - B) Interactive exploration of multidimensional cancer genomics data
   - C) Protein structure prediction
   - D) Social networking
   - **Answer: B**

9. **Which tab in cBioPortal provides a statistical analysis of alterations co-occurrence?**
   - A) OncoPrint
   - B) Mutual Exclusivity
   - C) Survival
   - D) Network
   - **Answer: B**

Note: This is a comprehensive set of MCQs covering the core concepts of Bioinformatics (BIF604) from the provided PDF. Good luck with your paper!